Chapter 1

Chess at Bell Labs

> When in doubt, use brute force.
> — Ken Thompson

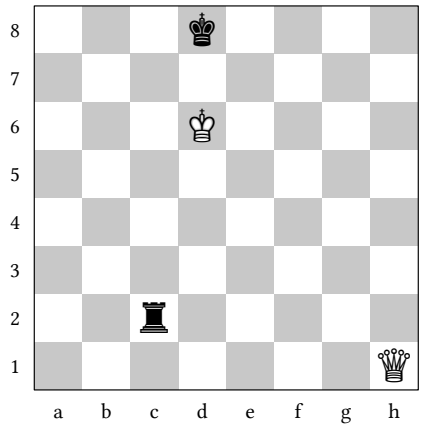
The building in Murray Hill where Unix was written, and where troff first set a page, also played chess. Ken Thompson and Joe Condon built Belle there, a machine whose move generator and evaluation were wired in hardware; it searched some hundred thousand positions a second, won the World Computer Chess Championship in 1980, and in 1983 became the first machine to earn a master's rating.

1.1 The endgame by retrograde analysis

Thompson's lasting contribution to chess was not Belle's play but its endgames. With few pieces on the board, every position can be listed. Beginning from the mates, a program works backwards: a position from which the side to move can reach a mate is won, one from which every move reaches a won position for the opponent is lost, and so on until nothing changes. What remains is drawn. The result is perfect play, and it overturned a good deal of received theory.

Queen against rook was among the first endings so solved. The queen wins, and every textbook says so; but the defence is stubborn, and the winning method long. When a grandmaster undertook to win such positions against the machine, the rook proved far harder to catch than he had expected, the machine finding at each move the defence that held out longest. Later databases of five pieces showed endings

long thought drawn to be won, in more moves than the rules then allowed, and the fifty-move rule was for a time given exceptions.



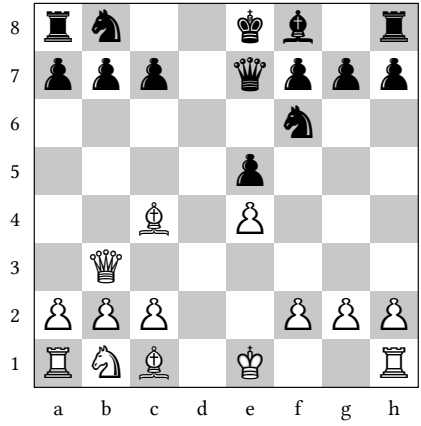
1.2 A game in the old style

Played in a box at the Paris opera in 1858, between Paul Morphy and two amateurs in consultation, the Duke of Brunswick and Count Isouard, it is the most often shown game of chess, for it shows in seventeen moves why development matters.

1.e4 e5 2.♘f3 d6 3.d4 ♗g4

The pin is premature; White can break it at once.

4.dxe5 ♗xf3 5.♕xf3 dxe5 6.♗c4 ♘f6 7.♕b3 ♕e7



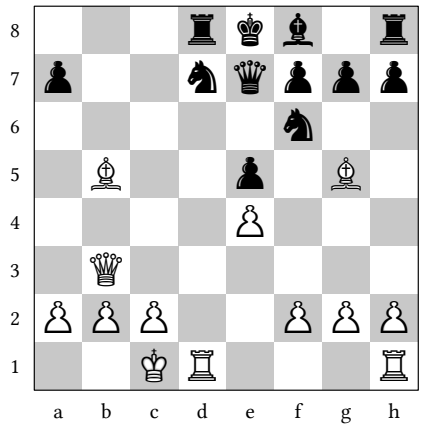
Black has surrendered the bishop pair and the centre. White's queen and bishop bear down on f7, and the queen on b7 as well.

8.♘c3

Morphy develops rather than take the pawn: 8.♕xb7 ♕b4+ would force the queens off.

8...c6 9.♗g5 b5 10.♘xb5 cxb5 11.♗xb5+ ♘bd7 12.0-0-0 ♖d8

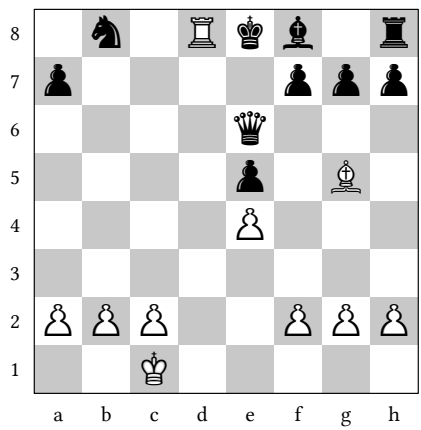
White has given a knight to open the lines; every piece is now in play, and Black's king is still in the centre.



13.♖xd7 ♖xd7 14.♖d1 ♕e6 15.♗xd7+ ♘xd7

White to move mates in two.

16.♕b8+! ♘xb8 17.♖d8#



The rook mates on d8, defended by the bishop on g5. White mates with only a rook and a bishop left, against Black's queen, rook, bishop and knight.